

FermiForge

Current two-dimensional quasiclassical solver

Equations, numerical algorithm, and code architecture

This note records the algorithm that is implemented now for a translation-invariant two-dimensional cross-section of a spin-triplet superfluid. Its immediate purpose is to make the normal-core, A-phase-core, and double-core vortex development auditable. It also identifies the interfaces that can later support walls, leads, discontinuous-Galerkin discretisation, and portable accelerator execution.

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The words *implemented*, *validation*, *experimental*, and *planned* are used deliberately. They distinguish working code from the intended architecture and avoid turning a roadmap statement into a claim about the present solver.

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1 Executive overview

The implemented calculation is a nonlinear fixed-point problem for the full complex triplet order-parameter tensor and a three-component, current-related Fermi-liquid mean field on a two-dimensional Cartesian cross-section. Every MPI rank currently holds the complete spatial state. Active target points are distributed among ranks; at each target the code integrates stable coherence functions along all quasiclassical directions and positive Ozaki poles, reconstructs the spin–Nambu propagator, and forms a new local mean field. A global reduction assembles the mapped field, after which rank zero projects selected zero modes and applies the source-compatible Anderson accelerator.

The central separation of concerns is

$$\begin{array}{c} \boxed{\text{spatial field}} \longrightarrow \boxed{\text{ray sampling}} \longrightarrow \boxed{\text{Riccati transport}} \\ \boxed{\text{local self-consistency map}} \longrightarrow \boxed{\text{nonlinear update}}. \end{array} \quad (1)$$

The field grid, the one-dimensional trajectory integration mesh, and the nonlinear iteration vector are therefore distinct objects. This is important both for multiscale accuracy and for a future GPU implementation.

Maturity at this snapshot

Implemented	Full complex 3×3 triplet field; full 2×2 spin coherence and 4×4 spin–Nambu propagator; static nonuniform rectilinear mesh; Q1 interpolation; straight rays; free-vortex asymptotic continuation; MPI point distribution; source-compatible Anderson mixing; checkpointing; optional zero-mode projection.
Validated	Unit and regression tests for mesh sampling, radial embedding, trajectory transport, propagator reconstruction, self-consistency integrands, MPI rank agreement, and normal-core radial/2D map comparison. Converged radial normal- and A-core fields are references; this does not by itself certify every modern 2D relaxation as converged.
Experimental	Free double-core relaxation, angle-dependent asymptotic state fitting, dependent-halo regeneration, staged domain studies, zero-mode removal, and held-out exterior “shadow” iteration.
Planned	Error-driven block AMR, discontinuous-Galerkin transport, container walls and reflection, interfaces/leads, dipole-energy self-consistency, physical observable normalisations, GPU offload, and distributed spatial fields.

The present implementation is therefore best described as an *MPI-enabled research prototype designed for later scaling*, not yet as a production large-cluster or GPU solver.

2 Provenance and notation

The implementation deliberately preserves several numerical conventions from the working radial code in `new_src`. The physical notation is aligned with the supplied double-core manuscript [1]; where a general quasiclassical convention and the historical source convention could differ, the code-compatible quantity is named explicitly rather than silently identified with a more general object.

The spatial coordinates are $\mathbf{r} = (x, y)$ and the vortex axis is z , with $\partial_z = 0$. Momentum remains three dimensional,

$$\hat{\mathbf{p}} = (p_x, p_y, p_z) = \left(\sqrt{1 - p_z^2} \cos \theta_p, \sqrt{1 - p_z^2} \sin \theta_p, p_z \right), \quad |\hat{\mathbf{p}}| = 1. \quad (2)$$

Greek indices $\alpha, \beta \in \{x, y, z\}$ label spin components and Roman indices $i, j \in \{x, y, z\}$ label orbital components. Repeated component indices are summed when this is unambiguous.

The code-basis triplet field is

$$A_{\alpha i}(\mathbf{r}) \in \mathbb{C}^{3 \times 3}, \quad d_{\alpha}(\hat{\mathbf{p}}, \mathbf{r}) = A_{\alpha i}(\mathbf{r}) p_i, \quad (3)$$

and the gap matrix is

$$\hat{\Delta}(\hat{\mathbf{p}}, \mathbf{r}) = \sum_{\alpha i} A_{\alpha i}(\mathbf{r}) i \sigma_{\alpha} \sigma_y p_i = (\mathbf{d} \cdot \boldsymbol{\sigma}) i \sigma_y = \begin{pmatrix} -d_x + i d_y & d_z \\ d_z & d_x + i d_y \end{pmatrix}. \quad (4)$$

The stored real vector $\mathbf{v}(\mathbf{r})$ is the coefficient used to construct the diagonal Fermi-liquid shift. It is called a *current-related mean field*; it is not yet a mass current in physical units. With the caller-supplied feedback scale a_{FL} ,

$$\nu(\hat{\mathbf{p}}, \mathbf{r}) = a_{\text{FL}} \hat{\mathbf{p}} \cdot \mathbf{v}(\mathbf{r}), \quad a_{\text{FL}} = \frac{F_1^s}{1 + F_1^s/3} \quad \text{in the present run configurations.} \quad (5)$$

At each mesh node the nonlinear state consequently has

$$9 \text{ complex } A_{\alpha i} + 3 \text{ real } v_i = 21 \text{ real unknowns.} \quad (6)$$

3 Quasiclassical transport problem

In conventional notation, the equilibrium propagator obeys an Eilenberger transport equation of the form

$$i \hbar \mathbf{v}_F \cdot \nabla \check{g} + [i \omega_n \check{\tau}_3 - \check{\Sigma}, \check{g}] = 0, \quad \check{g}^2 = -\mathbf{1}_4, \quad (7)$$

in the dimensionless normalisation used by the compatibility layer. The current code does not integrate the matrix equation directly. It evolves two stable Riccati/coherence branches from opposite ends of each ray and reconstructs $\check{g}$ at the target [3].

3.1 Projected ray and Q1 field sampling

For target $\mathbf{r}_0 = (x_0, y_0)$ and direction (2), the projected trajectory is

$$\mathbf{r}(s) = \mathbf{r}_0 + s(p_x, p_y), \quad \mathbf{r}(0) = \mathbf{r}_0. \quad (8)$$

The line is clipped to the rectangular storage mesh, the target is inserted exactly at $s = 0$, and every interval is bounded by a separate trajectory step control. A direction parallel to the invariant axis has one spatial sample, but still retains its full momentum in (3).

On a rectilinear cell $[x_0, x_1] \times [y_0, y_1]$, set

$$\xi = \frac{x - x_0}{x_1 - x_0}, \quad \eta = \frac{y - y_0}{y_1 - y_0}. \quad (9)$$

The interpolation operator is

$$Q_1[f](x, y) = (1 - \xi)(1 - \eta) f_{00} + \xi(1 - \eta) f_{10} \\ + (1 - \xi)\eta f_{01} + \xi\eta f_{11}. \quad (10)$$

It acts directly on every complex Cartesian component of A and every real component of $\mathbf{v}$; amplitudes and phases are not interpolated separately. The four point indices and weights form a reusable stencil. Thus fixed geometry can later be cached independently of changing field values.

3.2 Implemented Riccati coefficient equation

The exact compatibility kernel stores a coherence function as four complex coefficients $a = (a_0, \mathbf{a})$. Let $\alpha = (\alpha_0, \boldsymbol{\alpha})$ and $\beta = (\beta_0, \boldsymbol{\beta})$ be the local pair coefficients, $\zeta = iz_q - \nu$, and $\mathbf{h}$ the spin-vector diagonal self-energy. The transcribed ordinary differential equation is

$$\frac{da_0}{ds} = i \left[(a_0^2 + \mathbf{a} \cdot \mathbf{a}) \beta_0 + 2a_0(\zeta + \boldsymbol{\beta} \cdot \mathbf{a}) + \alpha_0 - 2\mathbf{h} \cdot \mathbf{a} \right], \quad (11)$$

$$\frac{da_j}{ds} = i \left[(a_0^2 - \mathbf{a} \cdot \mathbf{a}) \beta_j + 2a_j(\zeta + \boldsymbol{\beta} \cdot \mathbf{a} + \beta_0 a_0) + \alpha_j - 2h_j a_0 \right]. \quad (12)$$

For the active pure-triplet calculation,

$$a_0 = \alpha_0 = \beta_0 = 0, \quad \mathbf{h} = 0, \quad (13)$$

so that

$$\frac{d\mathbf{a}}{ds} = i [\boldsymbol{\alpha} + 2(\zeta + \boldsymbol{\beta} \cdot \mathbf{a})\mathbf{a} - (\mathbf{a} \cdot \mathbf{a})\boldsymbol{\beta}]. \quad (14)$$

On the forward branch $(\boldsymbol{\alpha}, \boldsymbol{\beta}) = (\mathbf{d}, \mathbf{d}^*)$; on the reverse branch $(\boldsymbol{\alpha}, \boldsymbol{\beta}) = (\mathbf{d}^*, \mathbf{d})$. These statements describe the implemented legacy-compatible branches. They do not yet define the most general equilibrium tilde operation for arbitrary singlet, triplet, exchange, and interface self-energies.

At an endpoint the source-compatible bulk value uses

$$\lambda = i\sqrt{\boldsymbol{\alpha} \cdot \boldsymbol{\beta} - \zeta^2}, \quad \mathbf{a} = -\frac{\boldsymbol{\alpha}}{\zeta + \lambda}, \quad (15)$$

with the Fortran complex square-root branch. An optional constant endpoint segment provides explicit relaxation before propagation through the sampled path.

Each interval is advanced by classical fourth-order Runge–Kutta while the self-energy coefficients are linearly interpolated to the RK stages. For Ozaki pole z_q , the interval substep count is

$$K_q = \max(K_{\min}, 1 + \text{int}(|z_q|)). \quad (16)$$

This reproduces the radial source arithmetic; it is independent of the maximum spacing used to construct the trajectory samples.

3.3 Spin–Nambu reconstruction and normalisation

Let $a^F = (a_0^F, a_x^F, a_y^F, a_z^F)$ and $a^R = (a_0^R, a_x^R, a_y^R, a_z^R)$ denote the coefficient sets returned by the forward and reverse integrations. The forward matrix is

$$\gamma = (a_0^F \mathbf{1}_2 + \mathbf{a}^F \cdot \boldsymbol{\sigma}) i \sigma_y = \begin{pmatrix} -a_x^F + i a_y^F & a_0^F + a_z^F \\ -a_0^F + a_z^F & a_x^F + i a_y^F \end{pmatrix}. \quad (17)$$

The reverse coefficients use the separate, tested sign map inherited from `new_src/riccati.f90`:

$$\tilde{\gamma}_{\text{src}} = (a_0^R \mathbf{1}_2 + a_x^R \sigma_x - a_y^R \sigma_y + a_z^R \sigma_z) i \sigma_y = \begin{pmatrix} -a_x^R - i a_y^R & a_0^R + a_z^R \\ -a_0^R + a_z^R & a_x^R - i a_y^R \end{pmatrix}. \quad (18)$$

Here $\mathbf{1}_2$ is the 2×2 identity in spin space. The subscript “src” is essential: $\tilde{\gamma}_{\text{src}}$ is the Nambu-partner matrix constructed from the legacy reverse branch, not the result of an already generalised tilde operation. An eventual field-level convention may take the form $\tilde{X}(\hat{\mathbf{p}}, \mathbf{R}, z) = X(-\hat{\mathbf{p}}, \mathbf{R}, -z^*)^*$, but that convention has not yet been frozen for every self-energy and interface object.

Define

$$N_+ = (\mathbf{1}_2 - \gamma \tilde{\gamma}_{\text{src}})^{-1}, \quad N_- = (\mathbf{1}_2 - \tilde{\gamma}_{\text{src}} \gamma)^{-1}. \quad (19)$$

The reconstructed blocks are

$$\begin{aligned} g &= -iN_+(\mathbf{1}_2 + \gamma\tilde{\gamma}_{\text{src}}), & f &= -2iN_+\gamma, \\ \tilde{f}_{\text{src}} &= 2iN_-\tilde{\gamma}_{\text{src}}, & g_h &= iN_-(\mathbf{1}_2 + \tilde{\gamma}_{\text{src}}\gamma). \end{aligned} \quad (20)$$

They form

$$\check{g} = \begin{pmatrix} g & f \\ \tilde{f}_{\text{src}} & g_h \end{pmatrix}. \quad (21)$$

Here g , f , $\tilde{f}_{\text{src}}$, and g_h are all 2×2 spin blocks, so $\check{g}$ acts in the four-dimensional spin–Nambu space. The pointwise diagnostic reported by the solver is exactly

$$\epsilon_{\text{norm}} = \max_{1 \leq \mu, \nu \leq 4} |(\check{g}^2 + \mathbf{1}_4)_{\mu\nu}|, \quad (22)$$

where $\mathbf{1}_4$ is the 4×4 identity. It is a maximum-entry error, not a Frobenius norm, and it tests the $\check{g}^2 = -\mathbf{1}_4$ code convention rather than the alternative $-\pi^2\mathbf{1}_4$ convention. Values near 10^{-15} in recent runs indicate algebraically consistent reconstruction to floating-point roundoff; they do not by themselves prove spatial or nonlinear convergence.

4 Quadratures and the local self-consistency map

The angular rule combines azimuthal midpoints with the supplied polar Gauss–Legendre nodes z_ℓ and weights w_ℓ^{GL} :

$$\theta_j = \frac{(j - \frac{1}{2})\pi}{N_\phi}, \quad (23)$$

$$\hat{\mathbf{p}}_{j\ell} = \left(\sqrt{1 - z_\ell^2} \cos \theta_j, \sqrt{1 - z_\ell^2} \sin \theta_j, z_\ell \right), \quad (24)$$

$$w_{j\ell} = \frac{w_\ell^{\text{GL}}}{2N_\phi}. \quad (25)$$

The half-circle azimuthal sampling is intentional: opposite propagation is represented by the two stable Riccati branches. It should not be documented as an independently sampled full 2π sphere.

The Ozaki table supplies temperature T , positive poles z_q , and residues R_q [4]. The spectral argument is iz_q and the current gap prefactor is

$$C_\Delta = \frac{T}{\log T + T \sum_q R_q / z_q}. \quad (26)$$

The transport temperature is therefore the value stored in the Ozaki table; the seed temperature is only an initialisation parameter.

For one direction and pole define the three anomalous combinations

$$s_x = \frac{1}{4} \left[f_{\downarrow\downarrow} - f_{\uparrow\uparrow} - (\tilde{f}_{\text{src},\downarrow\downarrow} - \tilde{f}_{\text{src},\uparrow\uparrow})^* \right], \quad (27)$$

$$s_y = -\frac{i}{4} \left[f_{\downarrow\downarrow} + f_{\uparrow\uparrow} - (\tilde{f}_{\text{src},\downarrow\downarrow} + \tilde{f}_{\text{src},\uparrow\uparrow})^* \right], \quad (28)$$

$$s_z = \frac{1}{4} \left[f_{\uparrow\downarrow} + f_{\downarrow\uparrow} - (\tilde{f}_{\text{src},\uparrow\downarrow} + \tilde{f}_{\text{src},\downarrow\uparrow})^* \right]. \quad (29)$$

The implemented local fixed-point map is then

$$\mathcal{F}_A[A, \mathbf{v}]_{\alpha i} = C_\Delta \sum_{j\ell q} 3w_{j\ell} R_q s_\alpha(\hat{\mathbf{p}}_{j\ell}, z_q) p_{j\ell, i}, \quad (30)$$

$$\mathcal{F}_v[A, \mathbf{v}]_i = \sum_{j\ell q} \frac{T}{2} w_{j\ell} R_q \text{Re Tr}_{\text{spin}} g(\hat{\mathbf{p}}_{j\ell}, z_q) p_{j\ell, i}. \quad (31)$$

Notice that C_Δ multiplies the gap map only. The field $\mathcal{F}_v$ is fed back through (5); a physical current observable requires a separately documented prefactor.

5 Spatial mesh, active domain, and free-vortex exterior

5.1 Static multiscale rectilinear mesh

The current 2D mesh is a nonuniform tensor-product grid with fine, medium, and coarse coordinate zones. It is useful for resolving a hard core, a broader soft core, and an outer sampling halo with fewer nodes than a uniform grid. It is *not* error-driven adaptive mesh refinement. The mesh and active mask remain fixed during one Anderson solve; a mesh change invalidates both cached interpolation stencils and the nonlinear history.

Every rank presently holds the full grid and state. Three distinct masks are important:

Mapped targets

Points for which the expensive quasiclassical map is evaluated. This can include diagnostic probes outside the update region.

Update points

Independent nonlinear unknowns packed into the Anderson vector. Pinned points and dependent exterior points are excluded.

Dependent halo

Stored mesh points regenerated from the asymptotic model after each accepted update. They remain available for Q1 trajectory sampling but are not independent Anderson degrees of freedom.

Thus “inactive” never means nonexistent. A point can be inactive for relaxation while still supplying trajectory data or an independent diagnostic.

5.2 Angle-dependent asymptotic continuation

For a singly quantised free vortex, the bulk reference is

$$A^{(0)}(\phi) = \Delta_B e^{i(N\phi + \phi_0)} R, \quad R \in SO(3). \quad (32)$$

The phase ϕ_0 and rotation R can be fitted from an annular/elliptical reference surface or supplied theoretically. Following Eqs. (19)–(20) of the double-core manuscript [1], the exterior field is represented as

$$A(r, \phi) = A^{(0)}(\phi) + A^{(1)}(\phi) \frac{R_c(\phi)}{r} + A^{(2)}(\phi) \left[\frac{R_c(\phi)}{r} \right]^2 + O(r^{-3}), \quad (33)$$

with the Cartesian block restriction

$$A^{(1)} = \begin{pmatrix} 0 & 0 & A_{xz} \\ 0 & 0 & A_{yz} \\ A_{zx} & A_{zy} & 0 \end{pmatrix}, \quad A^{(2)} = \begin{pmatrix} B_{xx} & B_{xy} & 0 \\ B_{yx} & B_{yy} & 0 \\ 0 & 0 & B_{zz} \end{pmatrix}. \quad (34)$$

The coefficient functions may vary with polar angle. Hence the continuation does not impose axial symmetry and can represent the double-core far field. Each order-parameter component has one permitted leading power; the code does not fit an arbitrary mixture of $1/r$ and $1/r^2$ to every component.

More explicitly, let $\rho = R_c(\phi)/r$, $\delta A_c = A(R_c(\phi), \phi) - A^{(0)}(\phi)$, and let P_1 select matrix entries with exactly one z index, with $P_2 = 1 - P_1$. The implemented state continuation is

$$A(r, \phi) = A^{(0)}(\phi) + \rho[P_1 \odot \delta A_c] + \rho^2[P_2 \odot \delta A_c]. \quad (35)$$

Here $\odot$ denotes componentwise multiplication. The order-parameter value on the inner fit surface is deliberately not used: continuity and the single power allowed by (34) fix each coefficient from the matching surface. Its angular dependence is inherited pointwise from δA_c .

The current-related mean field is continued as

$$\mathbf{v}(r, \phi) = \mathbf{v}^{(1)}(\phi) \frac{R_c(\phi)}{r} + \mathbf{v}^{(2)}(\phi) \left[\frac{R_c(\phi)}{r} \right]^2 + O(r^{-3}). \quad (36)$$

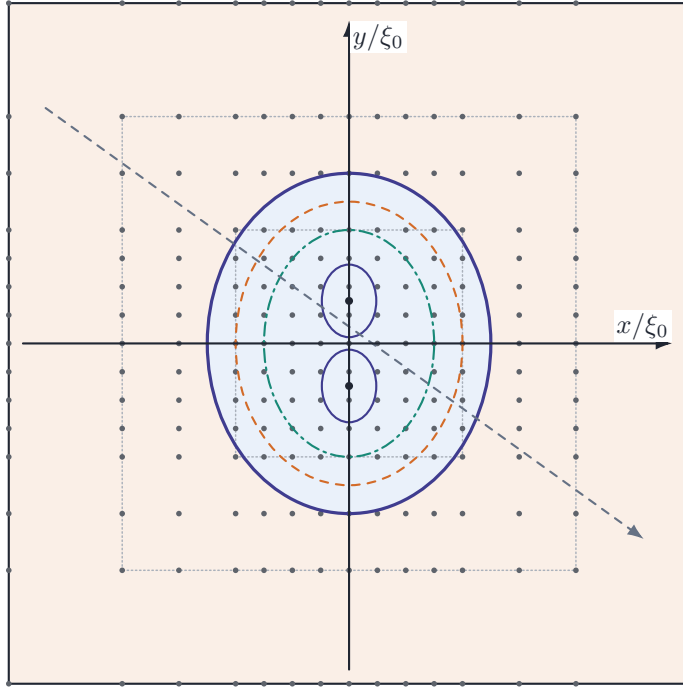
Two nested circular or elliptical source surfaces determine its two coefficients independently for each angle. The outer mesh halo is then regenerated from (33)–(36) after every accepted production update. Quasiclassical rays are further extended from the storage rectangle through this analytic state to an enclosing circle, where their stable endpoint values are initialised.

For completeness, with $q = R_c/R_{\text{in}}$, $\mathbf{v}_{\text{in}} = \mathbf{v}(R_{\text{in}}, \phi)$, and $\mathbf{v}_c = \mathbf{v}(R_c, \phi)$, the actual two-point coefficient solve is

$$\mathbf{c}_2 = \frac{\mathbf{v}_{\text{in}} - q\mathbf{v}_c}{q(q-1)}, \quad \mathbf{c}_1 = \mathbf{v}_c - \mathbf{c}_2, \quad \mathbf{v}(r, \phi) = \rho\mathbf{c}_1 + \rho^2\mathbf{c}_2. \quad (37)$$

Figures 1 and 2 summarise the present construction. The first uses a deliberately small multiscale cell so that every stored point and every change of spacing remains visible. Its dimensions are illustrative rather than production choices. The second unwraps one fixed- ϕ ray and separates coefficient extraction, halo construction, and independent exterior verification.

Small illustrative multiscale cell



How to read it

Stored nodes

$15 \times 15 = 225$ points on $[-6\xi_0, 6\xi_0]^2$. Each dot is one Q1 interpolation node.

Three spacing bands

$\Delta = 0.5\xi_0$ for $|x|, |y| \leq 2\xi_0$;
 $\Delta = 1\xi_0$ out to $4\xi_0$;
 $\Delta = 2\xi_0$ in the outer band.

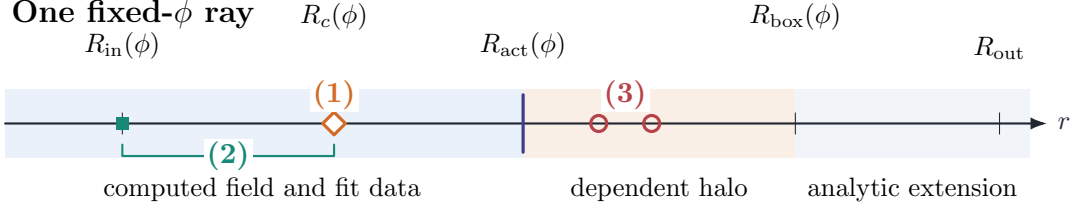
State regions

- $R_{\text{act}}(\phi)$: updated
- - - $R_c(\phi)$: matching
- · - · $R_{\text{in}}(\phi)$: inner fit

Inside R_{act} : independent Anderson variables. Outside: stored halo rebuilt from the fitted tail.

- - - $\rightarrow$ generic transport ray

Figure 1: Reduced mesh showing the multiscale-grid principle and the split between the independent update region and dependent asymptotic halo. All 225 displayed nodes occupy their stated coordinates, but the small cell and the ellipse sizes are illustrative rather than production-resolution choices. Line style and direct labels duplicate every colour distinction.



(1) Order parameter – one matching value at $R_c(\phi)$

$$\delta A_c(\phi) = A(R_c(\phi), \phi) - A^{(0)}(\phi),$$

$$A(r, \phi) = A^{(0)}(\phi) + \rho[P_1 \odot \delta A_c] + \rho^2[P_2 \odot \delta A_c], \quad \rho = R_c(\phi)/r.$$

(2) Current-related mean field – two source values

$$\{\mathbf{v}_{\text{in}}(\phi), \mathbf{v}_c(\phi)\} \longrightarrow \{\mathbf{c}_1(\phi), \mathbf{c}_2(\phi)\},$$

$$\mathbf{v}(r, \phi) = \rho \mathbf{c}_1(\phi) + \rho^2 \mathbf{c}_2(\phi).$$

(3) Independent exterior verification

Open-circle points are evaluated passively or relaxed in an isolated shadow solve. They never contribute to the fit and never change the production state.

R_{act} ends the independent update, R_{box} ends stored interpolation data, and R_{out} is the separate Riccati endpoint.

Figure 2: Asymptotic construction along one fixed- ϕ ray. Numbered markers identify the order-parameter input, the two mean-field inputs, and the held-out verification points described in the three cards. The five radii have distinct numerical roles; their displayed separations are not to scale.

The two-surface construction is the minimum identifiable fit for (36). A later overdetermined annular least-squares fit should add radial-window and angular-resolution diagnostics without relaxing the block constraint (34).

6 Nonlinear iteration

Let X denote the packed real state on the update mask and let

$$Y = \mathcal{F}(X), \quad r = Y - X \quad (38)$$

be the quasiclassical map and fixed-point residual. Packing contains the real and imaginary parts of all nine $A_{\alpha i}$ components followed by the three real mean-field components. A masked update preserves every non-update point exactly before the dependent halo is regenerated.

6.1 Residual diagnostics

For N packed values the code reports

$$e_{\text{avg}} = \frac{1}{N} \sum_{k=1}^N |r_k|, \quad e_{\text{rms}} = \sqrt{\frac{1}{N} \sum_{k=1}^N r_k^2}, \quad (39)$$

$$e_{\text{max}} = \max_k |r_k|, \quad e_{\text{rel}} = \frac{\|r\|_2}{\|X\|_2}. \quad (40)$$

It also records a legacy scaled maximum and the largest pointwise RMS over 21 values. Core and outer masks are reported separately so a large quiet region cannot hide a local defect. Reaching the iteration limit is a distinct terminal status and is not renamed convergence.

6.2 Zero-mode projection

The double-core solve may remove residual components parallel to numerically constructed tangents for global gauge, translations, and gauge-compensated orientation. A representative orientation tangent is

$$t_\theta[A] = iA + y \partial_x A - x \partial_y A, \quad (41)$$

with the corresponding spatial rotation of $\mathbf{v}$ but no gauge term. The tangents are orthogonalised, linearly dependent directions are rejected, and the removed norm and fraction are reported. Projection selects a numerical representative from a symmetry orbit; it does not change the physical self-consistency equations.

6.3 Source-compatible Anderson step

The modern `legacy_anderson_t` is a typed transcription of the working source iterator. It stores previous iterates X_i and residuals r_i , orders them by residual norm, and finds coefficients c_i that minimise

$$\left\| r_m + \sum_{i=1}^{m-1} c_i (r_i - r_m) \right\|_2^2. \quad (42)$$

Writing

$$d = \sum_{i=1}^{m-1} c_i (X_i - X_m), \quad (43)$$

$$r_{\min} = r_m + \sum_{i=1}^{m-1} c_i (r_i - r_m), \quad (44)$$

the accepted candidate is

$$X_{m+1} = X_m + d + p r_{\min}, \quad (45)$$

where the dynamically estimated p is clamped to $0.01 \leq p \leq p_{\max}$. The history length grows until its configured limit; this saturation is expected and does not mean that the residual-minimising search is disabled. Poor-progress tests can discard the largest-residual portion of the retained subspace. A singular subspace solve invokes the source-compatible deterministic sinusoidal fallback and is reported.

In the nonsingular branch the source-compatible mixing estimate is

$$\eta = \min \left(1, \sqrt{\frac{R_{\min}^{\text{previous}}}{\|r_{\text{current}}\|_2^2}} \right), \quad p = \text{clip}_{[0.01, p_{\max}]} \left\{ \frac{\eta}{2} \left(\frac{\|d\|_2}{\|r_{\text{base}}\|_2} + p_{\text{old}} \right) \right\}. \quad (46)$$

The first update is the simple step $X_1 = X_0 + 0.01 r_0$. The implemented stopping rule is the unnormalised condition $\max_k |r_k| \leq \varepsilon$; the other residual measures remain essential diagnostics.

Only rank zero owns the Anderson history and performs this update. The next typed field and its reports are broadcast to all ranks. Checkpoints contain the field, not the Anderson history; a restart therefore begins a fresh accelerator history.

7 End-to-end algorithm

Figure 3 shows the current production path. The outer loop is a fixed-mesh solve. Post-solve exterior validation is intentionally outside that loop and cannot modify the accepted production state.

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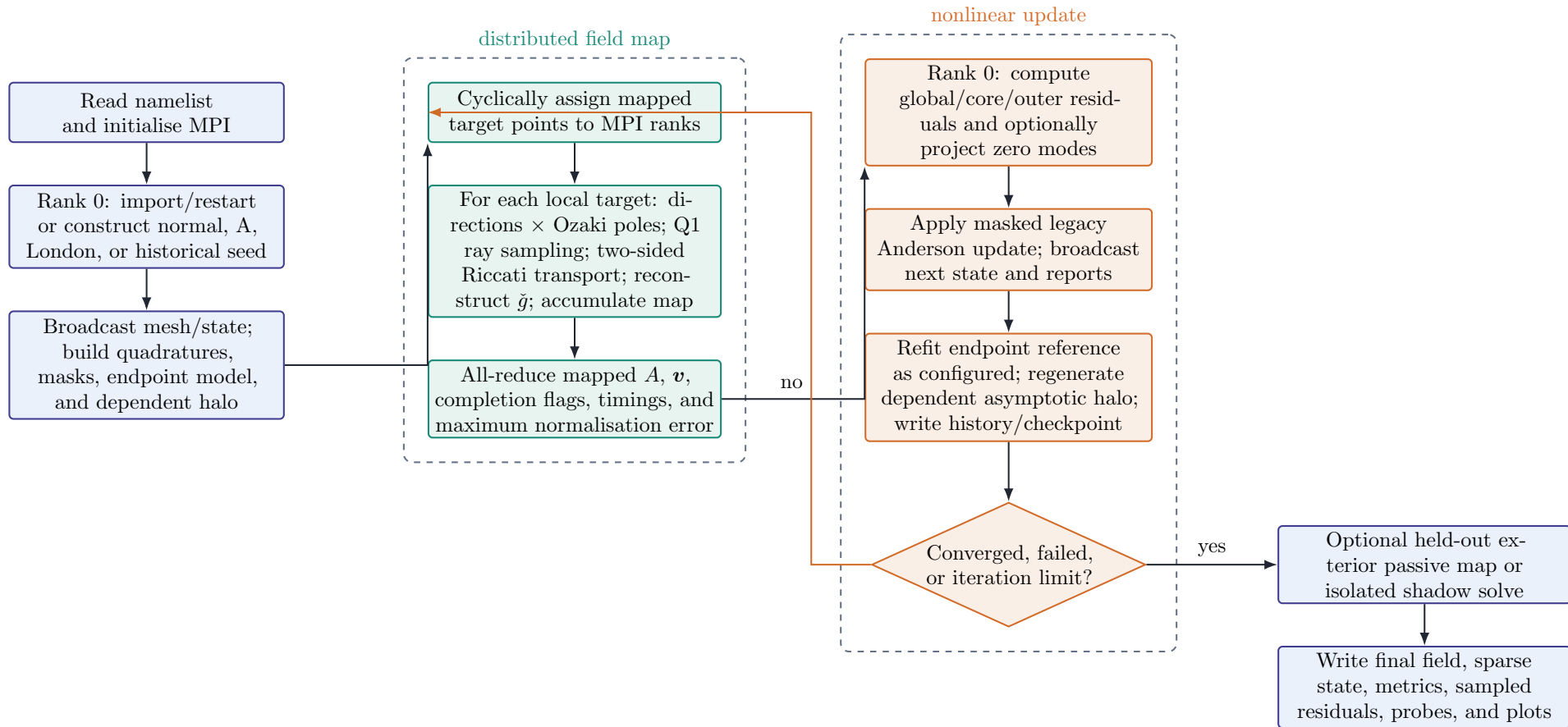


Figure 3: Implemented production algorithm. The expensive bracket is repeated for every nonlinear update. The independent exterior check is a validation stage and never feeds back into the production solution.

7.1 One mapped target point

For one target node a , the map can be expressed as the following nested algorithm:

1. For each angular quadrature direction, intersect the projected ray with the rectangular mesh, insert $s = 0$, limit path spacing, and cache Q1 stencils.
2. Sample A and v on the interior path. If the free-vortex endpoint is enabled, extend the path through the analytic exterior to its enclosing circle.
3. For each positive Ozaki pole, integrate the entry-stable and exit-stable coherence branches to the target using (11)–(16).
4. Convert both source coefficient sets to spin matrices and reconstruct $\tilde{g}$ using (20). Reject singular reconstruction; otherwise update (22).
5. Accumulate (30)–(31); after all directions and poles, apply C_Δ to the gap part only.

The natural unit of parallel work is therefore a target–direction–pole ray. The current CPU/MPI implementation distributes only target points, while the direction–pole loops remain available for later threaded or GPU batching.

8 Current code construction

The CMake build separates reusable non-MPI physics/data code in the `he3_mesh` library from MPI execution code in `he3_mpi_execution`. The double-core application source is exposed both under its benchmark name and as the user-facing `fermiforge_double_core_2d` executable. Figure 4 shows the principal call direction, not every Fortran `use` association.

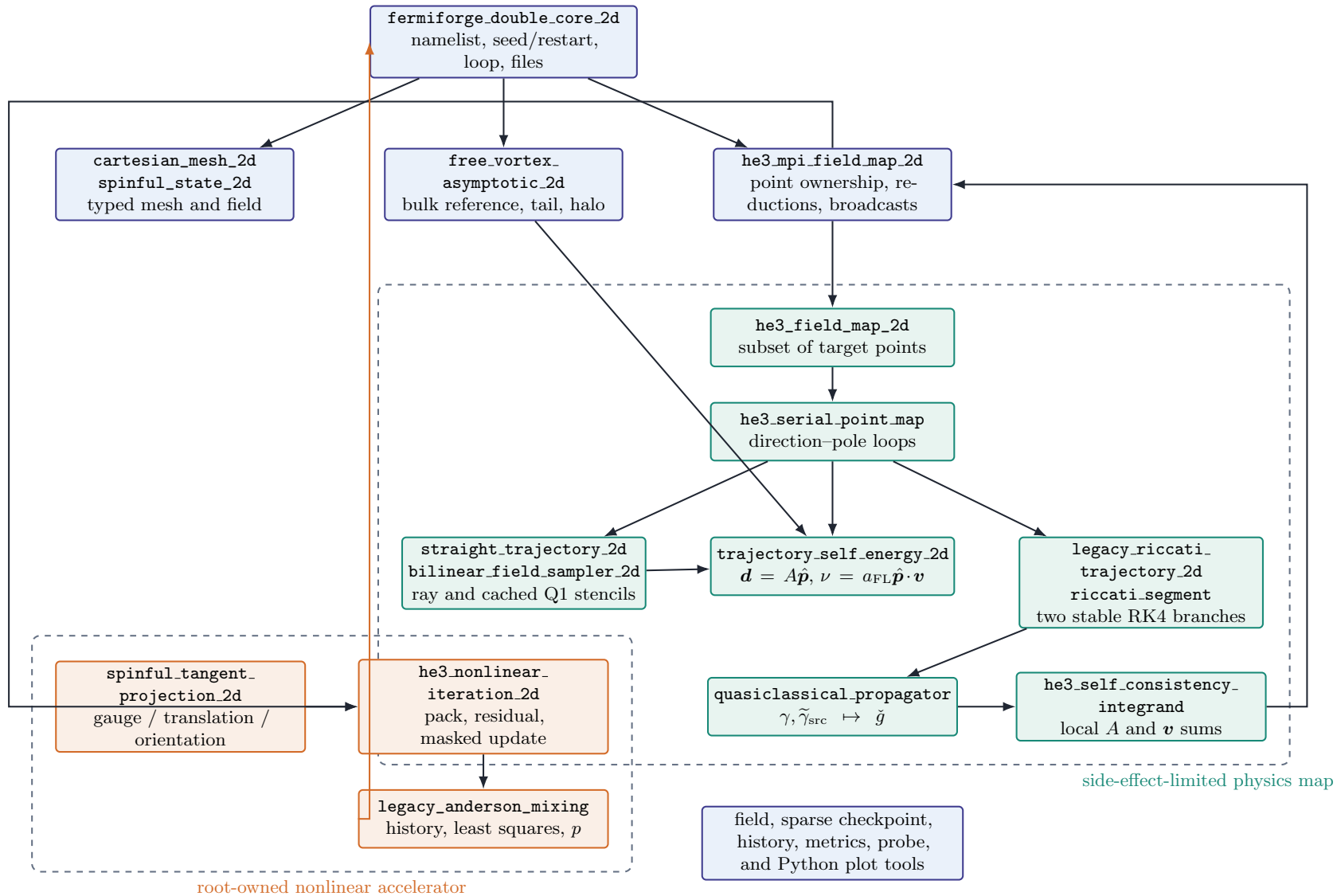


Figure 4: Principal implemented module pipeline. The reusable physics path is kept separate from MPI orchestration and from the application-level workflow.

8.1 MPI execution model

If active points are numbered $a = 0, \dots, N_a - 1$, rank r of P receives points satisfying $a \bmod P = r$. Each rank maps its local targets into a zero-initialised full-size output. `MPI_Allreduce` with a sum then assembles the unique active contributions; inactive points are copied exactly from the input state. Counts and timing sums are reduced, while the maximum normalisation error and rank elapsed time use a maximum reduction.

This replicated-state design is simple and already verifies equal numerical answers at multiple rank counts. It is not a spatial domain decomposition: there are no mesh-halo exchanges, and memory per rank grows with the full field. The first GPU stage can retain this model, batching direction-pole rays within each rank; larger meshes will eventually require a measured choice between distributed spatial blocks and direction/energy decomposition.

8.2 Checkpoint and output semantics

Dense field files contain coordinates, 18 real order-parameter entries, a pair-density proxy, and the three current-related mean-field components. Sparse checkpoints contain coordinates plus the 21 nonlinear values at a selected checkpoint mask. Restart overlays values onto exact mesh nodes and starts a new Anderson history.

Post-processing can transform the Cartesian 3×3 field to the frozen complex cylindrical/harmonic basis used by the radial source. These harmonic components are diagnostics and plotting products; the unconstrained 2D solver continues to store and interpolate the Cartesian tensor directly.

On an iteration-limited run, the final field/checkpoint contains the last *accepted* state, while the production residual describes the last evaluated pair $X \mapsto \mathcal{F}(X)$ before that acceptance. This distinction should be retained when comparing a plotted state with the last residual line.

9 Exterior verification and acceptance diagnostics

The asymptotic formula is part of the production boundary model, so evaluating it at the same points used to determine its coefficients is not an independent test. The implemented validation layer therefore adds exterior mesh points that are not production unknowns.

Let B be the frozen final production state on all ordinary points and let P denote a small mask of held-out exterior points. Two modes are available:

Passive defect Evaluate $\mathcal{F}(B) - B$ on P . This tests whether the extrapolated value is locally consistent with the microscopic map without changing it.

Shadow iteration

Keep every point outside P fixed, initialise an independent Anderson accelerator, and relax only P to a state S^* . The output stores B , the first map $\mathcal{F}(B)$, S^* , and the terminal map $\mathcal{F}(S^*)$.

The useful quantities are

$$\delta_{\text{passive}} = \mathcal{F}(B) - B, \quad (47)$$

$$\delta_{\text{correction}} = S^* - B, \quad (48)$$

$$\delta_{\text{terminal}} = \mathcal{F}(S^*) - S^*. \quad (49)$$

The correction is interpretable as an independently relaxed discrepancy only when the shadow solve reaches its stated convergence tolerance. A `shadow_not_converged` result is diagnostic data, not

a converged correction. The shadow state, history, and maps never alter production checkpoints, endpoint fits, or convergence status.

The minimum acceptance set for a physical double-core solution should include:

- nonlinear residuals resolved by hard-core, soft-core, and exterior zones;
- ϵ_{norm} from (22);
- mesh, active-domain, matching-surface, outer-circle, trajectory-step, direction, and Ozaki-pole convergence studies;
- stability of the half-core separation and relevant harmonics;
- passive and, when affordable, converged shadow exterior defects;
- agreement with radial normal- and A-core references in the symmetric limit and with archived double-core data when imported;
- symmetry checks and, before claiming energetic stability, a consistent free-energy comparison.

10 Observable and terminology safeguards

The plotted pair-density proxy is

$$n_{\text{pair}}^{\text{proxy}}(\mathbf{r}) = \frac{1}{3} \sum_{\alpha i} |A_{\alpha i}(\mathbf{r})|^2. \quad (50)$$

It is useful for locating half cores but is not the response-defined superfluid density. Likewise, $\mathbf{v}$ and (31) are current-related Fermi-liquid fields rather than a normalised physical mass current. A future observable layer must freeze energy and length units, field normalisation, and all physical prefactors before labels such as ρ_s or SI current density are used.

The implementation has the full complex triplet tensor and spin-matrix propagator, but “full spin” currently has defined limits: the diagonal spin-exchange self-energy is zero, singlet/mixed pairing is absent, and no spin-active interface boundary condition is connected. Those are extensions of the field/self-energy model, not merely larger arrays.

The historical arrays also do not by their type prove whether the stored code-basis tensor is exactly the manuscript’s A or its constant-rotation and bulk-amplitude-reduced $\tilde{A}$. Until that conversion and the code length/energy units are frozen by a documented regression, this note keeps the neutral name “code-basis A ” and avoids attaching an unverified unit symbol to every plotted coordinate.

11 Verification ladder

Development follows a ladder in which each new layer retains a lower-level oracle:

1. manufactured constant, affine, and bilinear fields test mesh lookup and Q1 interpolation;
2. radial harmonic conversion and Cartesian embedding test basis and phase conventions;
3. frozen trajectory checkpoints test the two legacy Riccati branches;
4. arbitrary nonsingular coherence matrices test reconstruction and (22);
5. a complete one-point map tests direction/pole accumulation;
6. radial versus 2D point maps test the spatial bridge;

7. one-, four-, and ten-rank runs test MPI invariance;
8. normal-core 2D relaxation tests the nonlinear loop;
9. A-phase-core runs stress the long $1/r$ soft tail;
10. double-core runs test broken axial symmetry, moving half cores, angle-dependent asymptotics, and zero modes;
11. held-out exterior probes test the asymptotic closure independently.

The seed family is also provenance-labelled. Radial normal/A solutions are reference embeddings; `nop`, `aop`, and `dop` reproduce the historical initialisation logic; the regularised London double-core seed is a smooth model seed, not the same object as historical `dop`; imported legacy 2D fields provide a later numerical oracle.

12 What is not yet implemented

Topic	Present boundary of the implementation
Spatial adaptation	Fine/medium/coarse tensor-product mesh and staged active ellipse changes exist. There is no quadtree, local error-driven refinement, two-to-one balancing, or automatic field transfer/restart.
Walls and devices	The general 2D solver has a free-vortex analytic exterior. Specular reflection exists only in historical work and is not connected to the modern 2D transport path. Leads, weak links, diffuse scattering, and spin-active interfaces are future work.
Long-scale physics	The asymptotic spin-rotation tail is represented, but a self-consistent dipole term and a multilevel texture solver are not connected.
Accelerators	Kernels and data flow are being separated for batching, but no OpenMP-target, OpenACC, CUDA, HIP, or SYCL offload path is implemented.
DG/FEM	The discontinuous-Galerkin approach of Seja and Löfwander [5] is a target architecture for general devices, not a feature of the present Q1-sampled ray solver. DG is a finite-element method, not a finite-difference method.
Scaling	MPI distributes target points with a replicated field. Functional rank agreement has been tested; production strong/weak scaling and multi-node memory limits have not yet been established.
Thermodynamics	A fully normalised physical current, superfluid-density tensor, free-energy functional, and stability comparison are not yet part of the modern output layer.

13 Development path from this architecture

The logical sequence is governed by verification gates rather than by replacing all infrastructure at once.

1. **Accept the free-vortex 2D reference.** Obtain reproducible normal-, A-, and double-core solutions; quantify domain, tail, quadrature, transport, and nonlinear convergence; compare against radial and archived 2D data.
2. **Turn staged multiscale studies into block adaptation.** Add dense rectangular blocks, conservative restriction/prolongation, local residual and gradient indicators, two-to-one balance, and solve–estimate–transfer–restart cycles. Rebuild ray stencils and reset Anderson whenever the mesh changes.

3. **Add a geometry and scattering interface.** Geometry supplies inside/outside tests, intersections, normals, and material identifiers; scattering supplies specular, diffuse, partially specular, or spin-active laws. For a specular wall with normal $\hat{\mathbf{n}}$,

$$\hat{\mathbf{p}}_{\text{out}} = \hat{\mathbf{p}}_{\text{in}} - 2(\hat{\mathbf{p}}_{\text{in}} \cdot \hat{\mathbf{n}})\hat{\mathbf{n}}, \quad (51)$$

while p_z is conserved for a wall extruded along the invariant axis.

4. **Evaluate the DG transition.** Use the validated point-map physics as an oracle while introducing the discontinuous-Galerkin weak form for general geometries. Decide polynomial order, flux/boundary treatment, and adaptivity from measured accuracy and cost rather than terminology.
5. **Batch the ray kernel and offload portably.** First build a structure-of-arrays batched CPU path. Then use a thin OpenMP-target execution layer as the initial vendor-neutral route, retaining identical physics kernels and a CPU reference. MPI remains the multi-node layer.
6. **Generalise material self-energies and observables.** Add singlet/triplet mixtures, exchange fields, impurity/interface terms, physical current/free energy, walls, weak links, and superconducting-magnetic devices behind explicit model interfaces.

This ordering preserves a trusted reference at each step: radial source $\rightarrow$ modern scalar CPU map $\rightarrow$ 2D MPI relaxation $\rightarrow$ adaptive/DG CPU solver $\rightarrow$ batched accelerator solver.

14 Reproducibility checklist for a run

A result should archive enough information to reconstruct both the physical map and the numerical approximation:

- Git revision and statement of any uncommitted changes;
- compiler, build type, MPI implementation, and rank count;
- complete namelist plus the exact Gauss and Ozaki tables;
- seed type or restart path and the bulk-reference policy;
- explicit x and y coordinates, update/mapped/checkpoint masks, and all circular/elliptical fit radii;
- trajectory step, Riccati substep floor, endpoint relaxation distance, outer circle, and asymptotic policy;
- Anderson history limit, p_{max} , projection policy, tolerance, and terminal status;
- iteration history, final accepted state, last evaluated residual, propagator-normalisation maximum, and shadow status;
- plotting scripts and colour-scale conventions used for the figures.

A Compact pseudocode

```

MPI initialise
rank 0: read input; import/restart or construct seed
broadcast mesh and full state
read angular and Ozaki rules; configure endpoint and masks
regenerate dependent asymptotic halo

```

```

for nonlinear iteration:
  distribute mapped target points cyclically over ranks
  for each local target point:
    accumulator = 0
    for each angular direction:
      build projected ray and cached Q1 stencils
      sample A and current-related field
      extend ray through analytic exterior when enabled
      for each positive Ozaki pole:
        integrate stable coherence from entry to target
        integrate stable reverse branch from exit to target
        reconstruct 4x4 propagator; check  $g^2 = -1$ 
        accumulate gap and mean-field contributions
      finalise target map
  all-reduce mapped state and diagnostics
  rank 0: compute residuals; optionally remove zero modes
  rank 0: Anderson update on update mask
  broadcast accepted state
  refit configured bulk reference; regenerate dependent halo
  write history/checkpoint; test stop condition

optional: freeze production state and evaluate/iterate held-out exterior probes
rank 0: write final fields, metrics, residuals, probes, plots
MPI finalise

```

B Principal source-to-module correspondence

Concern	Current module or application layer
Typed mesh and state	<code>cartesian_mesh_2d</code> , <code>spinful_state_2d</code>
Iteration packing	<code>new_src_iteration_layout_2d</code>
Q1 interpolation and rays	<code>bilinear_field_sampler_2d</code> , <code>straight_trajectory_2d</code>
Trajectory self-energy	<code>trajectory_self_energy_2d</code>
Riccati propagation	<code>riccati_segment</code> , <code>legacy_riccati_trajectory_2d</code>
Spin–Nambu reconstruction	<code>spin_matrix_2x2</code> , <code>quasiclassical_propagator</code>
Quadrature and point map	<code>he3_quadrature</code> , <code>he3_self_consistency_integrand</code> , <code>he3_serial_point_map</code>
Field and MPI maps	<code>he3_field_map_2d</code> , <code>he3_mpi_field_map_2d</code>
Asymptotic exterior	<code>free_vortex_asymptotic_2d</code>
Residual and Anderson update	<code>he3_nonlinear_iteration_2d</code> , <code>legacy_anderson_mixing</code>
Zero modes	<code>spinful_tangent_projection_2d</code>
Seeds	<code>double_core_seed_2d</code> , <code>historical_core_seed_2d</code> , ra- dial embedding/import modules
Application workflow	<code>app/benchmark_legacy_double_core_map_2d.f90</code> , built as <code>fermiforge_double_core_2d</code>
Run orchestration and plots	scripts in <code>tools/</code>

References

- [1] M. A. Silaev, E. V. Thuneberg, and M. Fogelström, “Structure, bound states, and rotational dynamics of the double-core vortex in superfluid $^3\text{He-B}$,” working manuscript, 11 September 2026. In particular, Eqs. (1), (19), and (20) supply the order-parameter and asymptotic conventions used here.
- [2] J. W. Serene and D. Rainer, “The quasiclassical approach to superfluid ^3He ,” *Physics Reports* **101**, 221–311 (1983), [doi:10.1016/0370-1573\(83\)90051-0](https://doi.org/10.1016/0370-1573(83)90051-0).
- [3] M. Eschrig, “Distribution functions in nonequilibrium theory of superconductivity and Andreev spectroscopy in unconventional superconductors,” *Phys. Rev. B* **61**, 9061 (2000), [doi:10.1103/PhysRevB.61.9061](https://doi.org/10.1103/PhysRevB.61.9061).
- [4] T. Ozaki, “Continued fraction representation of the Fermi–Dirac function for large-scale electronic structure calculations,” *Phys. Rev. B* **75**, 035123 (2007), [doi:10.1103/PhysRevB.75.035123](https://doi.org/10.1103/PhysRevB.75.035123).
- [5] K. M. Seja and T. Löfwander, “Finite element method for the quasiclassical theory of superconductivity,” *Phys. Rev. B* **106**, 144511 (2022), [doi:10.1103/PhysRevB.106.144511](https://doi.org/10.1103/PhysRevB.106.144511).